1. This question carries 40% of the mark.

(a) Consider each of the following continuous-time signals. For each case, specify if the signal is causal/non-causal, periodic/non-periodic, odd/even. If the signal is periodic specify its period.

(i) $x(t) = 2\cos(2\pi t)$ **[2]**

Answer

Non-causal, periodic, even

(ii) $x(t) = 5e^{2\text{sign}(t)\cdot t}$ **[2]**

Answer

Non-causal, non-periodic, even

(b) Consider the signal

$$x(t) = \begin{cases} 1-t, & 0 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Now sketch each of the following signals and describe briefly in words how each of the signals can be derived from the original signal $x(t)$.

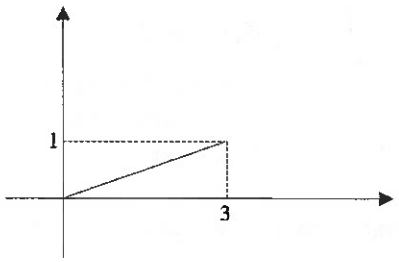
(i) $x\left(-\frac{t}{3}+1\right)$ **[2]**

(ii) $x(2t+5)$ **[2]**

Answer for (i) and (ii)

$$x\left(-\frac{t}{3}+1\right)$$

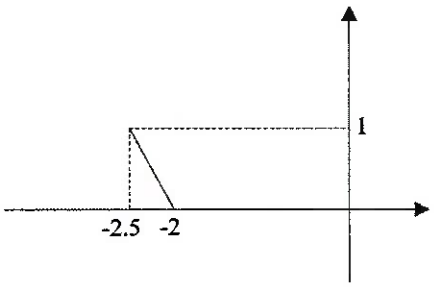
$$= \begin{cases} 1-\left(-\frac{t}{3}+1\right) = \frac{t}{3}, & 0 \le -\frac{t}{3}+1 \le 1 \Rightarrow -3 \le -t \le 0 \Rightarrow 0 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$$



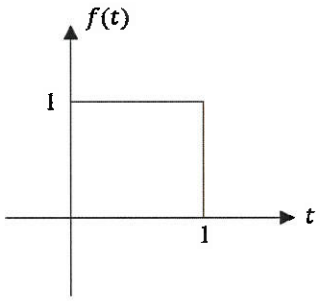
$$x(2t+5)$$

$$= \begin{cases} 1-(2t+5) = -2t-4, & 0 \le 2t+5 \le 1 \Rightarrow -5 \le 2t \le -4 \Rightarrow -\frac{5}{2} \le t \le -2 \\ 0 & \text{otherwise} \end{cases}$$

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(c) By applying the property of convolution, find the system's output (i.e., zero-state response) given that the impulse response is $h(t) = e^{-t}u(t)$ and that the input $f(t)$ is as shown in Figure 1(c) below.



[6]

Figure 1(c)

Answer

$h(t) = e^{-t}u(t)$

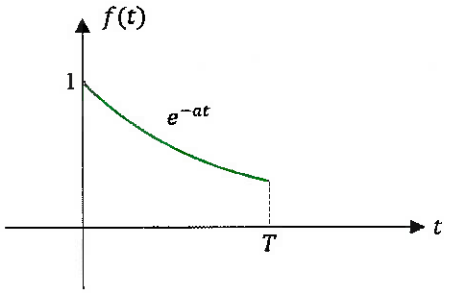
$f(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$

$y(t) = h(t) * f(t) = \int_{-\infty}^{\infty} h(\tau)f(t-\tau)d\tau \Rightarrow y(t) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau)f(t-\tau)d\tau$

$f(t-\tau) = \begin{cases} 1, & 0 \le t-\tau \le 1 \Rightarrow -1 \le \tau - t \le 0 \Rightarrow t-1 \le \tau \le t \\ 0 & \text{otherwise} \end{cases}$

$y(t) = \int_0^t e^{-\tau}d\tau = 1 - e^{-t}\ y(t), t \ge 0$

(d) Derive the Fourier transform of the signals $f(t)$ shown in Figures 1(d)(i)-(ii) below.
(i)



[3]

Figure 1(d)(i)

Answer

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$$F(\omega) = \int_0^T e^{-at}e^{-j\omega t}dt = \int_0^T e^{-(a+j\omega)t}dt = \frac{1 - e^{-(a+j\omega)T}}{a + j\omega}$$

(ii)

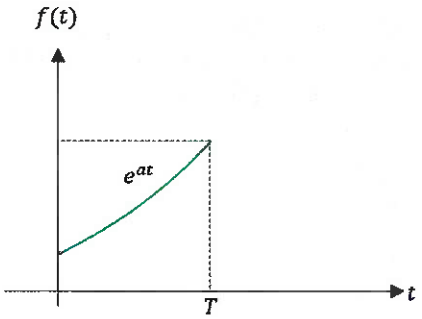


Figure 1(d)(ii)

[3]

Answer

$$F(\omega) = \int_0^T e^{at}e^{-j\omega t}dt = \int_0^T e^{(a-j\omega)t}dt = \frac{1 - e^{(a-j\omega)T}}{-a + j\omega}$$

(e) The output $y(t)$ of a continuous-time LTI system is related to the input $x(t)$ by the differential equation

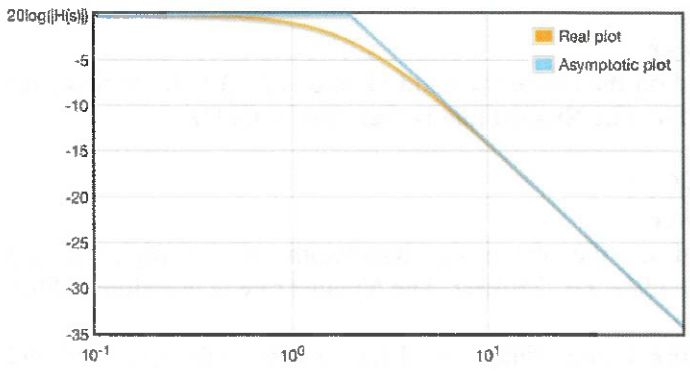
$$\frac{dy(t)}{dt} + 2y(t) = 2x(t)$$

Determine the frequency response of the system and sketch the asymptotic behavior of its Bode plots. **[5]**

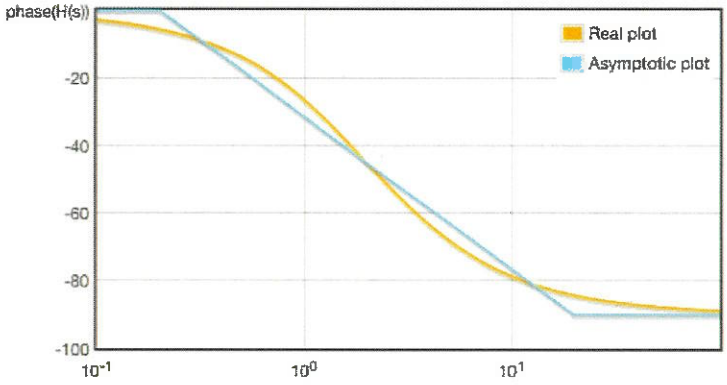
Answer

The frequency response is obtained by applying the Fourier Transform in both sides of the equation $\frac{dy(t)}{dt} + 2y(t) = 2x(t)$, i.e.,

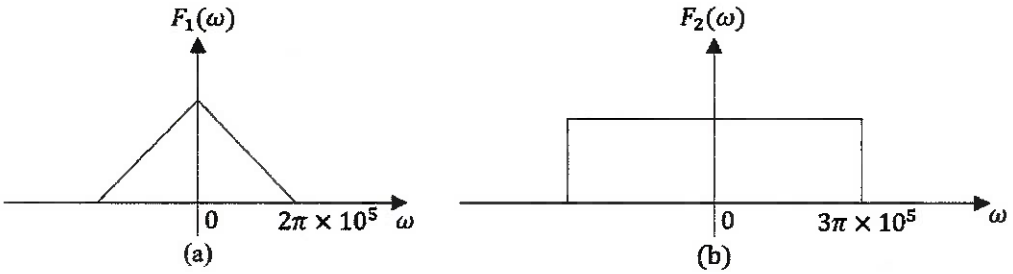
$Y(\omega)\, j\omega + 2Y(\omega) = X(\omega) \Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{2+j\omega}$



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(f) Figures (a) and (b) below show the Fourier spectra of signals $f_1(t)$ and $f_2(t)$. Determine the Nyquist sampling rates for the following signals. (Hint: Use the frequency convolution and the width property of the convolution.)



(i) $f_1(t)$ **[2]**

Answer

Bandwidth of $f_1(t)$ is 100kHz and Nyquist rate is 200kHz.

(ii) $f_2(t)$ **[2]**

Answer

Bandwidth of $f_2(t)$ is 150kHz and Nyquist rate is 300kHz.

(iii) $f_1^2(t)$ **[2]**

Answer

The Fourier Transform of $f_1^2(t)$ is $\frac{1}{2\pi}F_1(\omega) * F_2(\omega)$. From the property regarding the size of convolution we have that the bandwidth of $f_1^2(t)$ is twice the bandwidth of $f_1(t)$, i.e., 200kHz. The Nyquist rate is therefore, 400kHz.

(iv) $f_2^3(t)$ **[2]**

Answer

Based on the above the bandwidth of $f_2^3(t)$ is three times the bandwidth of $f_1(t)$, i.e., 450kHz. The Nyquist rate is therefore, 900kHz.

(v) $f_1(t)f_2(t)$ **[2]**

Answer

Based on the above the bandwidth of $f_1(t)f_2(t)$ is the sum of the individual bandwidths, i.e., 250kHz. The Nyquist rate is therefore, 500kHz.

(g) Consider the discrete-time, causal LTI system with input $x[n]$ and output $y[n]$ related with the difference equation:

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$$y[n]-\frac{5}{6}y[n-1]+\frac{1}{6}y[n-2]=x[n]-\frac{1}{2}x[n-1]$$

(i) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the impulse response of the above system.
[Hint: Use the fact that the z-transform $\frac{z}{z-a}$ corresponds to the function $a^n u[n]$ if $|z|>|a|$ and the function $-a^n u[-n-1]$ if $|z|<|a|$. The function $u[n]$ is the discrete-time unit step function.]
[3]

(ii) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the output if $x[n]=\left(\frac{1}{2}\right)^n u[n]$. **[2]**

Answer

(i)

$$Y(z)-\frac{5}{6}z^{-1}Y(z)+\frac{1}{6}z^{-2}Y(z)=X(z)-\frac{1}{2}z^{-1}X(z)$$
$$Y(z)(1-\frac{5}{6}z^{-1}Y(z)+\frac{1}{6}z^{-2})=X(z)(1-\frac{1}{2}z^{-1})$$
$$Y(z)(1-\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})=X(z)(1-\frac{1}{2}z^{-1})\Rightarrow$$
$$Y(z)\left(1-\frac{1}{3}z^{-1}\right)=X(z)\Rightarrow\frac{Y(z)}{X(z)}=\frac{1}{\left(1-\frac{1}{3}z^{-1}\right)}\Rightarrow H(z)=\frac{z}{z-1/3}$$

$h[n]=\left(\frac{1}{3}\right)^n u[n]$, $|z|>1/3$, causal system

$h[n]=-\left(\frac{1}{3}\right)^n u[-n-1]$, $|z|<1/3$

(ii)

$X(z)=\frac{z}{z-1/2}$, $|z|>1/2$

$Y(z)=\frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$, ROC: $|z|>\frac{1}{2}\cap|z|>\frac{1}{3}=|z|>\frac{1}{3}$

2. This question carries 30% of the mark.

(a) (i) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time, causal signal $x(t)=e^{-at}u(t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[3]**

Answer

Consider the signal $x(t)=e^{-at}u(t)$. The Fourier transform $X(j\omega)$ converges only for $a>0$ as shown in the following.

$$X(j\omega)=\int_{-\infty}^{+\infty}e^{-at}u(t)e^{-j\omega t}dt=\int_{0}^{+\infty}e^{-at}e^{-j\omega t}dt=\frac{1}{-(j\omega+a)}e^{-(j\omega+a)t}\Big|_0^{+\infty}=\frac{1}{j\omega+a},\ a>0$$

The Laplace transform is

$$X(s)=\int_{-\infty}^{+\infty}e^{-at}u(t)e^{-st}dt=\int_{0}^{+\infty}e^{-(s+a)t}dt$$

With $s=\sigma+j\omega$ we have

$$X(\sigma+j\omega)=\int_{0}^{+\infty}e^{-(\sigma+a)t}e^{-j\omega t}dt$$

The above is the Fourier transform of $e^{-(\sigma+a)t}u(t)$, and as shown above

$$X(\sigma+j\omega)=\frac{1}{(\sigma+a)+j\omega},\ \sigma+a>0$$

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or since $s=\sigma+j\omega$ and $\sigma=\text{Re}\{s\}$, we have

$$X(s)=\frac{1}{s+a},\ \text{ROC: Re}\{s\}>-a$$

(ii) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time, anti-causal signal $x(t)=-e^{-at}u(-t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[3]**

Answer

Consider the signal $x(t)=-e^{-at}u(-t)$. The Fourier transform $X(j\omega)$ converges only for $a<0$ as shown in the following.

$$X(j\omega)=\int_{-\infty}^{+\infty}-e^{-at}u(-t)e^{-j\omega t}dt=\int_{-\infty}^{0}-e^{-at}e^{-j\omega t}dt=\frac{1}{j\omega+a}e^{-(j\omega+a)t}\Big|_{-\infty}^{0}=\frac{1}{j\omega+a},\ a<0$$

The Laplace transform is

$$X(s)=\int_{-\infty}^{+\infty}-e^{-at}u(-t)e^{-st}dt=-\int_{-\infty}^{0}e^{-(s+a)t}dt$$

With $s=\sigma+j\omega$ we have

$$X(\sigma+j\omega)=-\int_{-\infty}^{0}e^{-(\sigma+a)t}e^{-j\omega t}dt$$

The above is the Fourier transform of $-e^{-(\sigma+a)t}u(-t)$, and thus,

$$X(\sigma+j\omega)=\frac{1}{(\sigma+a)+j\omega},\ \sigma+a<0$$

or since $s=\sigma+j\omega$ and $\sigma=\text{Re}\{s\}$, we have

$$X(s)=\frac{1}{s+a},\ \text{ROC: Re}\{s\}<-a$$

(iii) Show that the Laplace transform of the signal $\delta(t)$ is 1 with $\delta(t)$ the continuous impulse function. **[3]**

Answer

$$X(j\omega)=\int_{-\infty}^{\infty}\delta(t)e^{-j\omega t}dt=e^{-j\omega\cdot 0}=1$$

(b) Consider a continuous-time causal signal $x(t)$, $x(t)=0$, $t<0$ with Laplace transform $X(s)=\frac{s+2}{s-2}$. Find the analytical expression of $x(t)$, using the results obtained in question (a) above and also find the region of convergence of $X(s)$.
[Hint: you may write the above Laplace transform as $X(s)=1+\frac{4}{s-2}$]. **[4]**

Answer

$$X(s)=1+\frac{4}{s+2}\Rightarrow$$
$$x(t)=\delta(t)+4e^{2t}$$

(c) Consider a continuous Linear Time-Invariant system with input signal $x(t)=e^{-t}u(t)$, where $u(t)$ is the unit step function and output signal $y(t)=\frac{1}{3}e^{2t}u(-t)+\frac{1}{3}e^{-t}u(t)$.

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(i) Determine the transfer function of the system $H(s)$ and its Region of Convergence (ROC) knowing that the ROC of $Y(s)$ has to be the intersection of the ROCs of $X(s)$ and $H(s)$. **[5]**

Answer

$y(t) = \frac{1}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t)$

$Y(s) = -\frac{1}{3}\frac{1}{s-2} + \frac{1}{3}\frac{1}{s+1}$, ROC: $\text{Re}\{s\} < 2 \cap \text{Re}\{s\} > -1$

$x(t) = e^{-t}u(t) \Rightarrow X(s) = \frac{1}{s+1}$, ROC: $\text{Re}\{s\} > -1$

$H(s) = \frac{Y(s)}{X(s)} = -\frac{1}{3}\frac{s+1}{s-2} + \frac{1}{3} = \frac{-1}{s-2}$, $\text{Re}\{s\} < 2$

(ii) Determine the impulse response of the system $h(t)$. **[5]**

Answer

From the given information we deduct that the impulse response is anti-causal. Therefore, $h(t) = e^{2t}u(-t)$.

(d) The output $y(t)$ of an LTI system is related to the input $x(t)$ by the differential equation

$$\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} - 2y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of the system's impulse response $h(t)$.

(i) Determine $H(s)$ as a ratio of two polynomials. **[3]**

Answer

$(s^2 + s - 2)Y(s) = sX(s) + 2X(s) \Rightarrow (s+2)(s+1)Y(s) = (s+2)X(s) \Longrightarrow$

$\frac{Y(s)}{X(s)} = \frac{1}{s-1} = H(s)$

(ii) Determine $h(t)$ for each of the following cases:

1. The system is stable.
2. The system is causal.

[4]

Answer

$h(t) = e^{t}u(t)$, ROC: $\text{Re}\{s\} > 1$, stable

$h(t) = e^{t}u(t)$, ROC: $\text{Re}\{s\} > 1$, causal

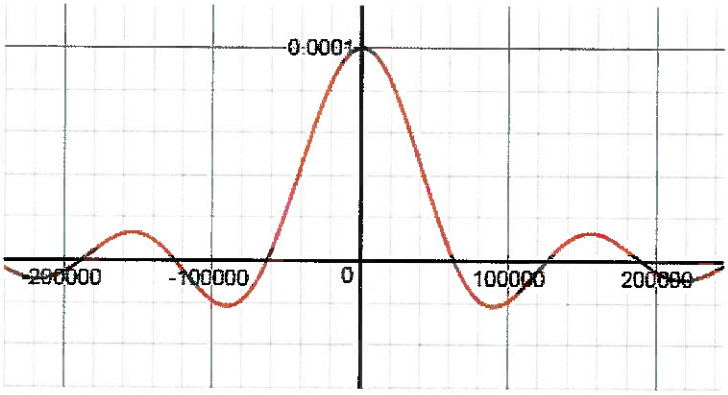
3. This question carries 30% of the mark.

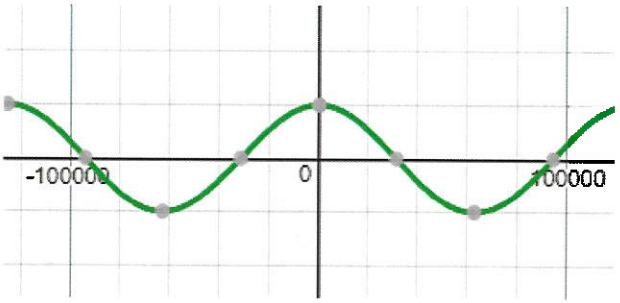
(a) Signals $f_1(t) = \Pi(10^4 t)$ and $f_2(t) = \frac{1}{2}(\delta(t - 0.2 \cdot 10^{-4}t) + \delta(t + 0.2 \cdot 10^{-4}t))$ are applied at the inputs of ideal lowpass filters $H_1(t) = 10^4\Pi\left(\frac{\omega}{40000\pi}\right)$ and $H_2(t) = \Pi\left(\frac{\omega}{20000\pi}\right)$. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$ as shown in Figure 3a below.

(i) Sketch $F_1(\omega)$ and $F_2(\omega)$. **[2]**

Answer

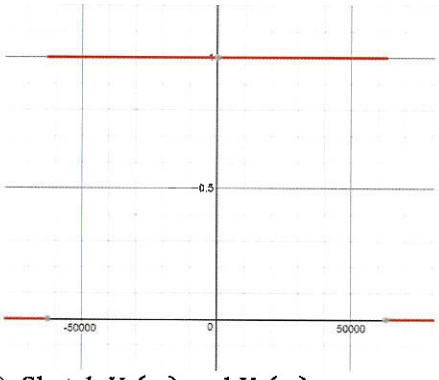
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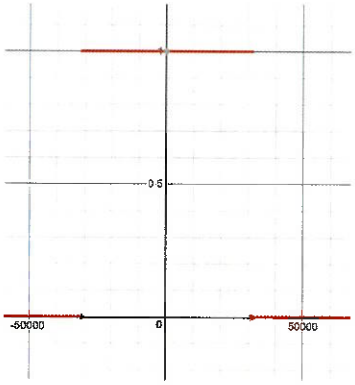




(ii) Sketch $H_1(\omega)$ and $H_2(\omega)$. **[2]**

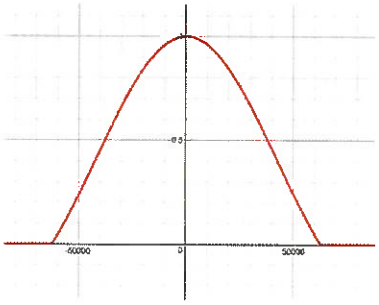
Answer

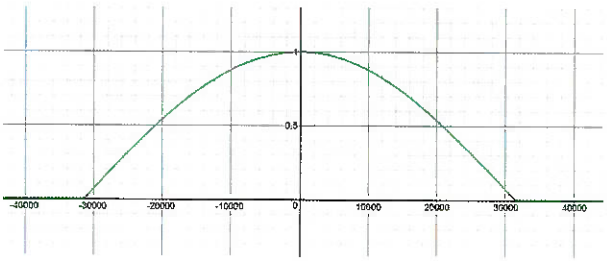




(iii) Sketch $Y_1(\omega)$ and $Y_2(\omega)$. **[2]**

Answer





(iv) Find the Nyquist sampling rate of $y_1(t)$, $y_2(t)$ and $y(t)$. **[2]**

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Answer
The bandwidth of $y_1(t)$ is 10kHz.
The bandwidth of $y_2(t)$ is 5kHz.
The bandwidth of $y_1(t)y_2(t)$ is 15kHz. The Nyquist rate is 30kHz.

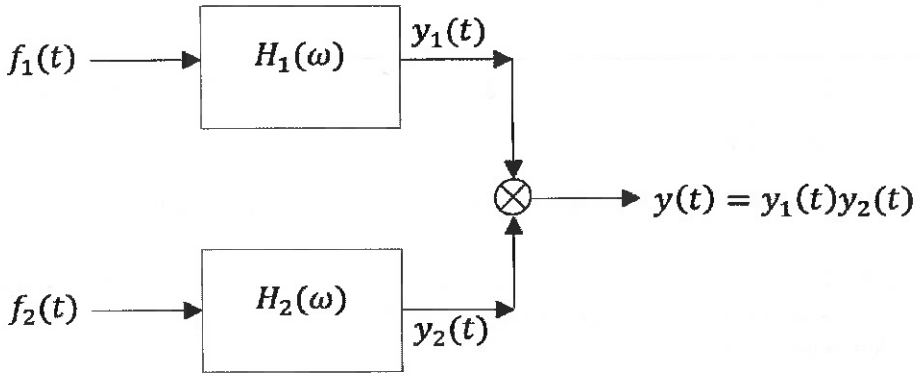


Figure 3a

(b) (i) Consider a discrete signal $x[n]$ with z-transform $X(z)$. Find the z-transforms of the signals $x_1[n] = x[n+3]$ and $x_2[n] = x[-n+1]$ as functions of the z-transform $X(z)$. **[5]**

Answer

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n+3]z^{-n} = \sum_{n=-\infty}^{\infty} x[n+3]z^{-(n+3)}\,z^3$$

$$X_2(z) = \sum_{n=-\infty}^{\infty} x_2[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n+1]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n+1]\left(\frac{1}{z}\right)^n \left(\frac{1}{z}\right)^{-1}\frac{1}{z}$$

$$= \frac{1}{z}\sum_{n=-\infty}^{\infty} x[-n+1]\left(\frac{1}{z}\right)^{-(-n+1)} = \frac{1}{z}\sum_{k=\infty}^{-\infty} x[k]\left(\frac{1}{z}\right)^{-k} = \frac{1}{z}X\left(\frac{1}{z}\right)$$

(ii) Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \text{ and } x_2[n] = \left(\frac{1}{3}\right)^n u[n]$$

with $u[n]$ the discrete unit step function. Given that

$$Z\{a^n u[n]\} = \frac{1}{1-az^{-1}},\ |z| > a$$

use the results obtained in question (b) (i) above to determine the z-transform $Y(z)$ of $y[n]$. **[5]**

Answer

$$Z\{x_1[n+3]\} = z^3 X_1(z) = z^3\frac{1}{1-\frac{1}{2}z^{-1}} = z^3\frac{z}{z-\frac{1}{2}} = z^4\frac{1}{z-\frac{1}{2}}$$

$$Z\{x_2[-n+1]\} = \frac{1}{z}X_2\left(\frac{1}{z}\right) = \frac{1}{z(1-\frac{1}{3}z)}$$

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$$Y(z) = Z\{x_1[n+3]\} \cdot Z\{x_2[-n+1]\} = z^4 \frac{1}{\left(z-\frac{1}{2}\right)} \cdot \frac{1}{z\left(1-\frac{1}{3}z\right)}$$

$$= -6z^3 \frac{1}{(2z-1)} \cdot \frac{1}{(z-3)}$$

(c) Consider an even sequence $x[n]$ (i.e., $x[n] = x[-n]$) with rational z-transform $X(z)$.

(i) From the definition of the z-transform, show that $X(z) = X\left(\frac{1}{z}\right)$. **[3]**

Answer

$g[n] = x[-n]$

$G(z) = \sum_{n=-\infty}^{\infty} g[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n]\, z^{n} = \sum_{n=-\infty}^{\infty} x[n]\left(\frac{1}{z}\right)^{-n}$

$= X\left(\frac{1}{z}\right)$

Therefore, it is obvious that if $x[n] = x[-n]$ then $X(z) = X\left(\frac{1}{z}\right)$.

(ii) From your result in part (c) (i), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$. **[3]**

Answer

Assume that $x[n] = x[-n]$ which implies $X(z) = X\left(\frac{1}{z}\right)$.

Assume that

$X(z) = P(z)(z - z_0)$

$X\left(\frac{1}{z}\right) = P\left(\frac{1}{z}\right)\left(\frac{1}{z} - z_0\right) = P\left(\frac{1}{z}\right)(1 - zz_0)\frac{1}{z}$

For $z = \frac{1}{z_0}$, $X\left(\frac{1}{z}\right) = 0 \Rightarrow \frac{1}{z_0}$ is a zero for $X\left(\frac{1}{z}\right)$.

Verify the results in part (c) (ii) for each of the following sequences:

(iii) $\delta[n+1] + \delta[n-1]$ **[3]**

Answer

$Z\{\delta[n+1]\} = z$

$Z\{\delta[n-1]\} = z^{-1}$

$$X(z) = Z\{\delta[n+1]\} + Z\{\delta[n-1]\} = z + z^{-1} = \frac{z^2+1}{z} = \frac{z^2 - i^2}{z} = \frac{(z+i)(z-i)}{z}$$

$$X\left(\frac{1}{z}\right) = \frac{(\frac{1}{z}+i)(\frac{1}{z}-i)}{\frac{1}{z}} = \frac{(1+zi)(1-zi)}{z} = \frac{(i-z)(i+z)}{iiz} = \frac{(z+i)(z-i)}{z}$$

The zeros are $+i, -i$. We know that $\frac{1}{i} = -i$ and $\frac{1}{-i} = i$. Therefore, $\frac{1}{i}$ and $\frac{1}{-i}$ are also zeros. The pole is 0. We see that $z = \infty$ also makes the z-transform ∞.

(iv) $\delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$ with $\delta[n]$ the discrete impulse function. **[3]**

$$x[n] = \delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$$

$$X(z) = z + z^{-1} - \frac{5}{2} = \frac{2z^2 + 2 - 5z}{2z} = \frac{(2z-1)(z-2)}{2z}$$

$$X\left(\frac{1}{z}\right) = \frac{1}{z} + (\frac{1}{z})^{-1} - \frac{5}{2} = \frac{1}{z} + z - \frac{5}{2} = \frac{2z^2 + 2 - 5z}{2z} = \frac{(2z-1)(z-2)}{2z}$$

Zeros are at 2 and ½ therefore their inverses are ½ and 2.

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